1/21/2022

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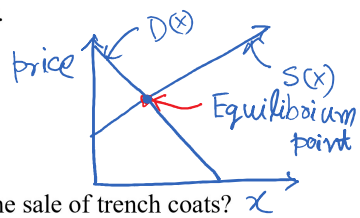
M168_Notes_1.4

1.4 Intersection of Straight Lines

Equilibrium Point

Is there a price that will satisfy the consumer and the producer? Plot the supply and demand equations on the same graph. If the lines intersect, that will be at a price and number of items that the producer and consumer agree upon. It is called the *equilibrium point*.



The equilibrium point is (x_0, p_0).

x_0 is the equilibrium quantity

p_0 is the equilibrium price.

Example: What is the equilibrium point for the sale of trench coats?

$\boxed{S(x) = D(x)}$

$2x + 60 = -2x + 200 \rightarrow 4x = 140$

$\Rightarrow x = 35$

$\boxed{x = 35}$, price $= 200 - 2 \cdot 35 = 130$

$\boxed{(35, 130)}$

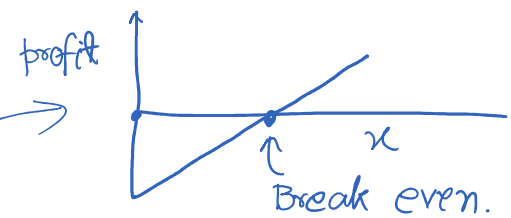
Yes!

$S(35) = D(35)$

Break-Even Analysis

Where the company's costs are equal to their revenue is the break-even point.

Example - What is the break-even point for the company making heaters?



$P(x) = R(x) - C(x)$

$= 50x - (40x + 1100)$

$= 10x - 1100 = 0$

$\Rightarrow 10(x - 110) = 0$

$\Rightarrow x = 110$

1.5 The Method of Least Squares

The table below shows the demand for boxes of Lunchie treats where x is the number of boxes (in thousands) and y is the price in dollars.

x	11	16	21	27	32
y	4.00	3.15	3.25	3.00	2.75

Graph this in a scatter plot. What might the demand equation look like?

With linear regression we want to find a line that takes all of the data into account. The line should come as close as possible to all the data points. The line that has the smallest sum of the distance from all the points is the least squares (or regression) line. The equation of the line is found using your calculator or excel – do not use the formulas in the book!

The correlation coefficient, r measures how close the data points are to the line. The closer the value is to $|1|$, the better the linear fit is. If the value is near 0, the data is not very linear.

Example: Use linear regression to find the demand equation for Lunchie treats. How many boxes would be sold at a price of \$3.75 each? If the company wants to see 14,000 boxes, what should the price be?

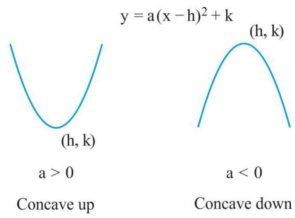
Quadratic Models

A quadratic is a polynomial of order 2:

$$q(x) = ax^2 + bx + c,\ a \neq 0.$$

Every quadratic function can be written in standard form:

$$y = a(x-h)^2 + k \text{ where } h = -\frac{b}{2a} \text{ and } k = c - \frac{b^2}{4a}$$



Example: What is the revenue from selling espressos if the demand equation for selling espressos is $p = -0.25 + 5$? Graph the revenue equation and interpret the result.